ANIMAL CELL TECHNOLOGY: PRIMARY CELL ISOLATION FROM SHRIMP HEPATOPANCREAS

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Abstract

This study presents a standardized laboratory workflow for the isolation and establishment of a primary cell suspension from shrimp hepatopancreatic tissue. The procedure integrated mechanical mincing with enzymatic dissociation using 0.25% Trypsin in Leibovitz L-15 medium, followed by sequential filtration through 100 μm and 40 μm mesh to ensure optimal single-cell recovery. The resulting suspension was concentrated via centrifugation at 500g for 10 minutes. Cell viability and density were subsequently quantified using the Trypan blue exclusion assay in a 1:1 dilution ratio via hemocytometer analysis. The protocol successfully yielded a viable primary cell suspension, demonstrating that the synergy of mechanical and enzymatic treatments is an effective approach for harvesting crustacean cells. These findings provide a foundational methodology for downstream applications in marine biotechnology, pathobiology, and in vitro cellular assays.

Keywords: Shrimp; Primary cell culture; Hepatopancreas; Enzymatic dissociation; Trypan blue exclusion; Cell viability.

1. Introduction

In animal cell technology, the successful establishment of a primary cell culture is a critical foundational step. To equip students with essential knowledge and techniques for in vitro cellular research, this laboratory exercise guides them through the complete workflow of isolating a primary cell suspension. This process encompasses tissue dissection, mechanical dissociation, enzymatic digestion, sequential filtration, and ultimately, cell viability quantification using crustacean tissue—specifically, the shrimp pancreas. The chosen environment for maintaining the cells during the dissociation process was Leibovitz's L-15 medium. This formulation is widely used for maintaining cells in standard atmospheric environments without the need for specialized incubators. For the active breakdown of the tissue, a 0.25% Trypsin solution was utilized.

2. Materials and Methods

2.1. Experimental Workflow

The experimental procedure for primary cell isolation is illustrated in Figure 2.1. The methodology integrated mechanical dissociation followed by enzymatic treatment to yield a cell suspension for further analysis [1, 3].

2.2. Biological Material

Fresh shrimp specimens were procured from a local market and transported to the laboratory under controlled conditions. The study was con-

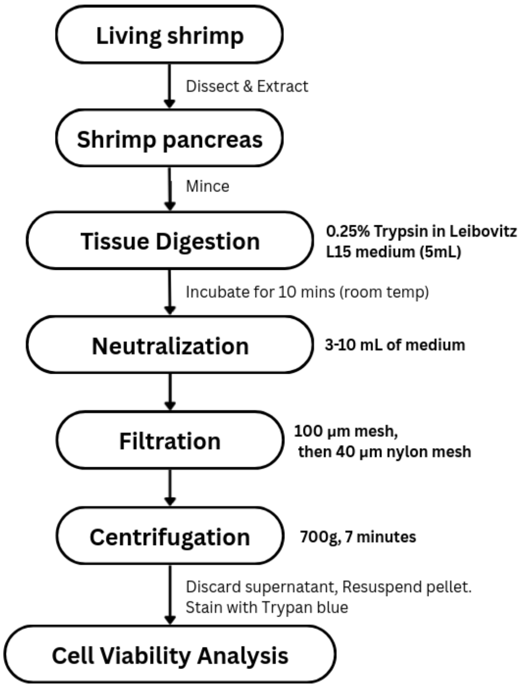

Figure 1: Workflow of primary cell isolation from shrimp hepatopancreas.

ducted at the Molecular Biology Laboratory, Department of Biotechnology, Ho Chi Minh City University of Technology (HCMUT).

2.3. Reagents and Instrumentation

The essential reagents and equipment utilized in this protocol are detailed in Table 1. All solutions were prepared or maintained at their optimal working concentrations [2, 3].

Table 1: Key Reagents and Equipment

Category	Specification / Concentration
Basal Medium	Leibovitz L-15 Medium [2]
Dissociation Agent	0.25% Trypsin-EDTA [3]
Filtration System	100 μm and 40 μm Nylon Mesh
Vital Stain	0.4% Trypan Blue Solution [5]
Centrifugation	Digital Microcentrifuge

2.4. Tissue Processing and Enzymatic Digestion

Aseptic surgical techniques were employed to excise the hepatopancreas from the shrimp. The isolated tissue was mechanically minced into fragments (< 1 mm^3) to maximize the surface area-to-volume ratio [3]. Subsequently, the tissue fragments were suspended in 5 mL of Leibovitz L-15 medium supplemented with 0.25% Trypsin-EDTA. Enzymatic dissociation was performed at 25°C for 10 minutes with periodic gentle agitation.

2.5. Filtration and Concentration

Post-incubation, the enzymatic activity was neutralized by the addition of 3–10 mL of fresh L-15 medium. The resulting homogenate was passed through a sequential filtration system (100 μm followed by 40 μm) to isolate a single-cell suspension. The filtrate was collected in sterile tubes and centrifuged at 500 g for 10 minutes at room temperature.

2.6. Resuspension and Viability Assessment

The supernatant was aspirated, and the cell pellet was carefully resuspended in 1 mL of L-15 medium. Cell density and viability were quantified using a hemocytometer following 1:1 dilution with Trypan Blue [4, 5].

3. Results & Discussion

The cell isolation from shrimp hepatopancreas was quantified using a hemocytometer after a 1:1 dilution. The results revealed a total count of 299 cells across 5 squares, with 55 live cells identified (Table 2).

Table 2: Hemocytometer Counts and Viability Analysis

Square	Total (T)	Live (L)	Dead (D)	Viability (%)
Top	61	15	46	24.6%
Left	46	12	34	26.1%
Center	64	14	50	21.9%
Right	64	6	58	9.4%
Bottom	64	8	56	12.5%
Sum/Avg	299	55	244	18.4%

3.1. Calculations

1. **Percentage Viability:**

$$\%V = \frac{\sum L}{\sum T} \times 100 = \frac{55}{299} \times 100 \approx 18.4\%$$

2. **Total Cell Concentration:**
(Including live and dead cells)

$$\text{Cells/mL} = \frac{\sum T}{5} \times DF \times 10^4 = 1.196 \times 10^6$$

3. **Live Cell Concentration:**
(DF = 2 due to 1:1 mix)

$$\text{Cells/mL} = \frac{\sum L}{5} \times DF \times 10^4 = 2.2 \times 10^5$$

4. **Total Yield (1 mL):**

$$\text{Yield} = 2.2 \times 10^5 \text{ viable cells}$$

The data demonstrates a successful isolation of cells, but with a low viability of 18.4%, which is significantly below the typical 80% benchmark for healthy primary cultures [3, 6]. The high mortality rate suggests that the 0.25% Trypsin-EDTA treatment (10 min) may have been too aggressive for the sensitive shrimp hepatopancreatic tissue, or that the mechanical dissociation introduced excessive physical stress. The resulting concentration of 2.2×10^5 cells/mL provides a sufficient population for initial analysis, but the low survival rate indicates a need for protocol optimization, such as reducing enzymatic concentration or incubation time to preserve cellular integrity [1].

4. Conclusion

The protocol involving trypsinization and sequential filtration (100→40 μm) successfully isolated shrimp hepatopancreatic cells with a concentration of 2.2×10^5 cells/mL. However, the low viability (18.4%) highlights the sensitivity of crustacean primary cells to standard enzymatic dissociation. Future studies should focus on optimizing dissociation parameters to improve cellular survival and establish more robust primary culture models.

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